

POLITECNICO
MILANO 1863

DEPARTMENT OF MECHANICAL
ENGINEERING

EXTRA EXERCISES

EXERCISE CLASS 6

ARIMA

Name Surname



DIPARTIMENTO DI ECCELLENZA
MIUR 2018-2022



Milano

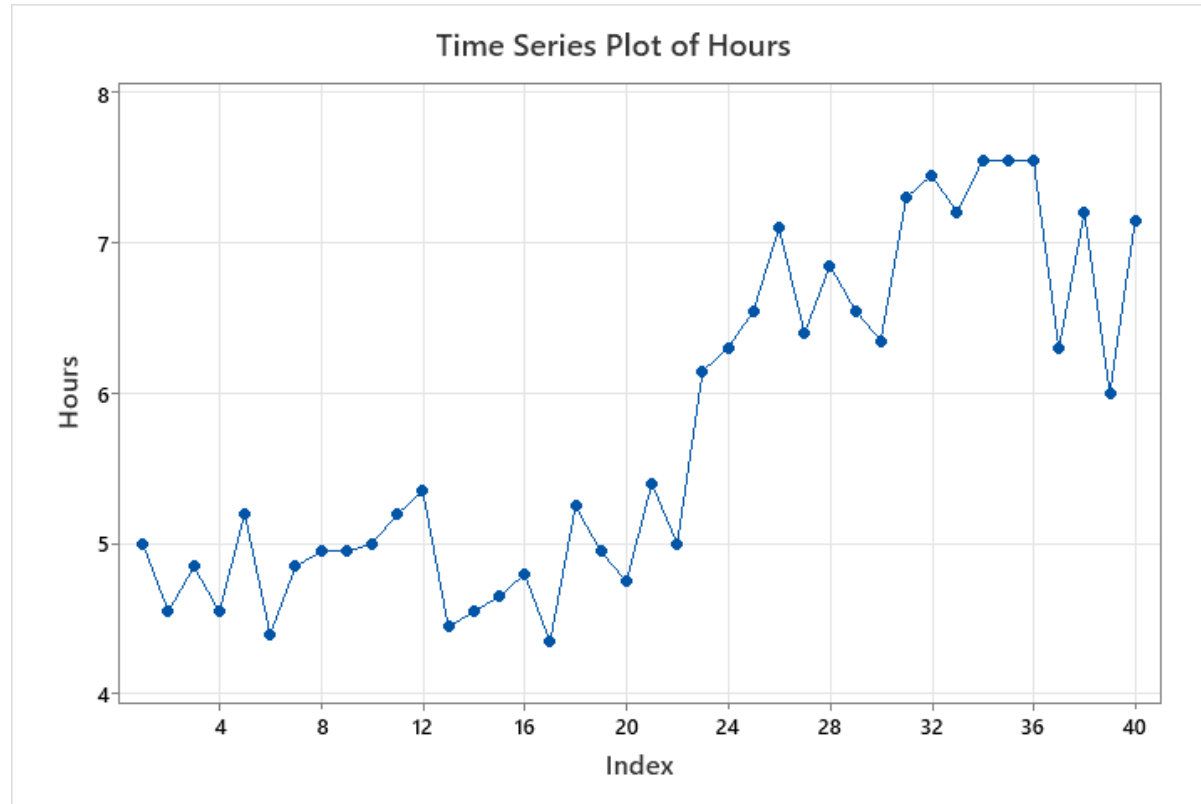
EXTRA EXERCISE 1

The number of hours required to complete an assembly operation of a vehicle is monitored for each unit. The data is reported in the file '*hours.csv*'.

1. Inspect the dataset and verify the assumptions.
2. Suggest a suitable model for the data.

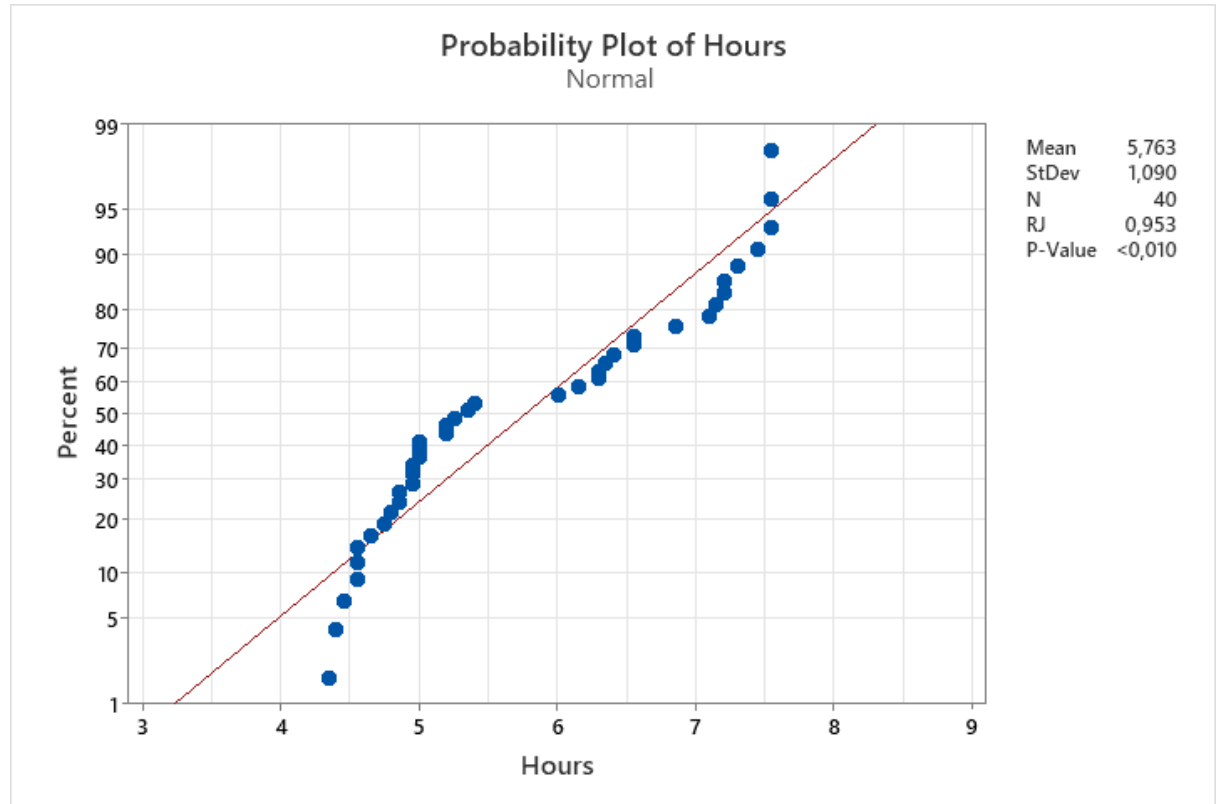
Point 1

Inspect the dataset and verify the assumptions.



It looks like the data are either following a trend or a meandering pattern.

Normality assumption:

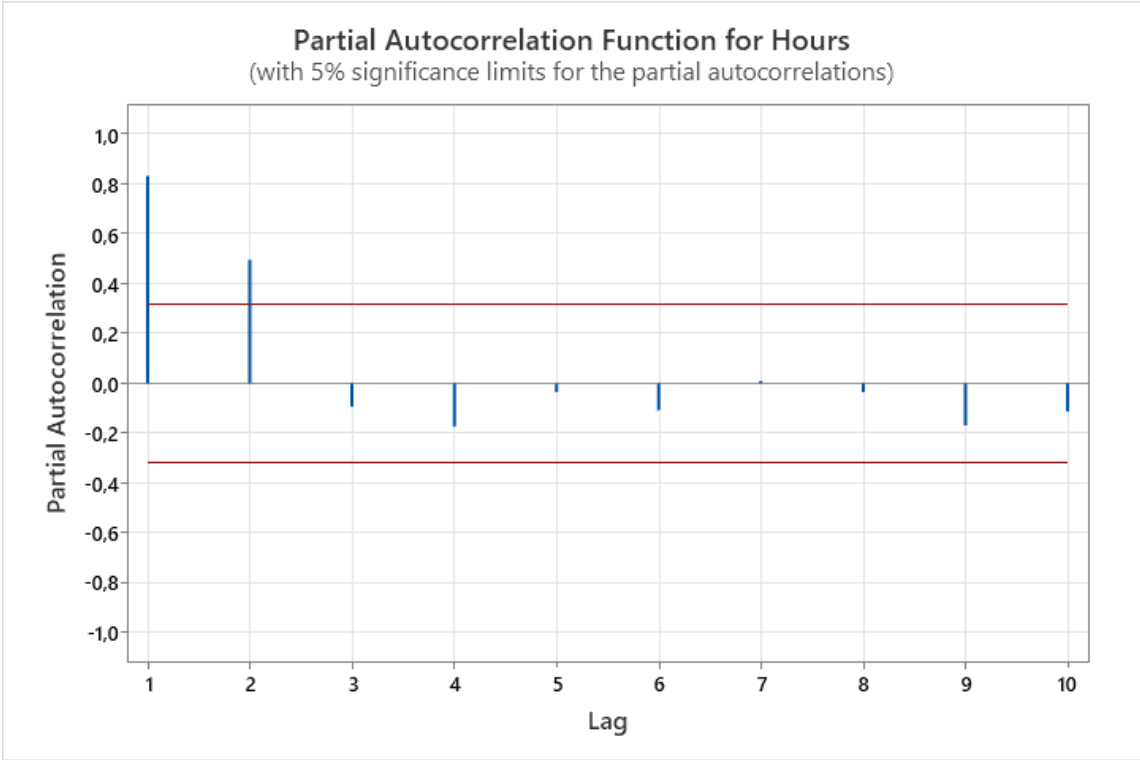
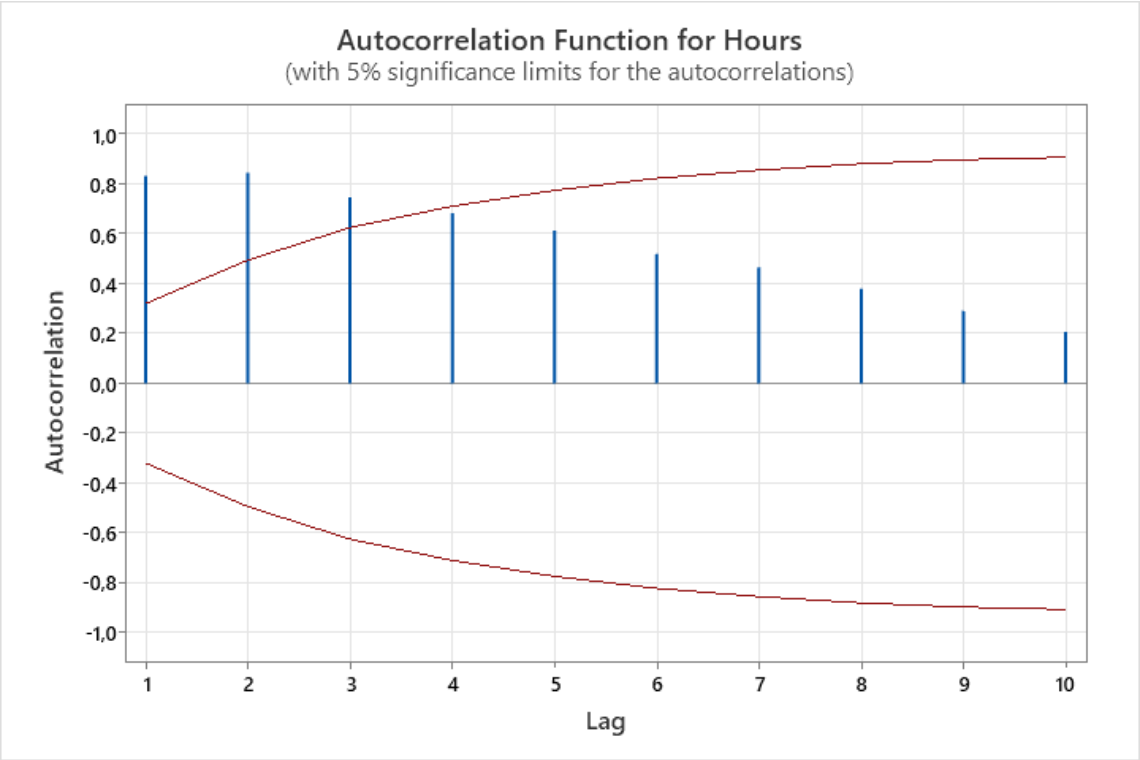


p-value < 0.01, hence we **reject** the normality assumption

Point 1

Inspect the dataset and verify the assumptions.

Independence assumption:



Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Runs test:

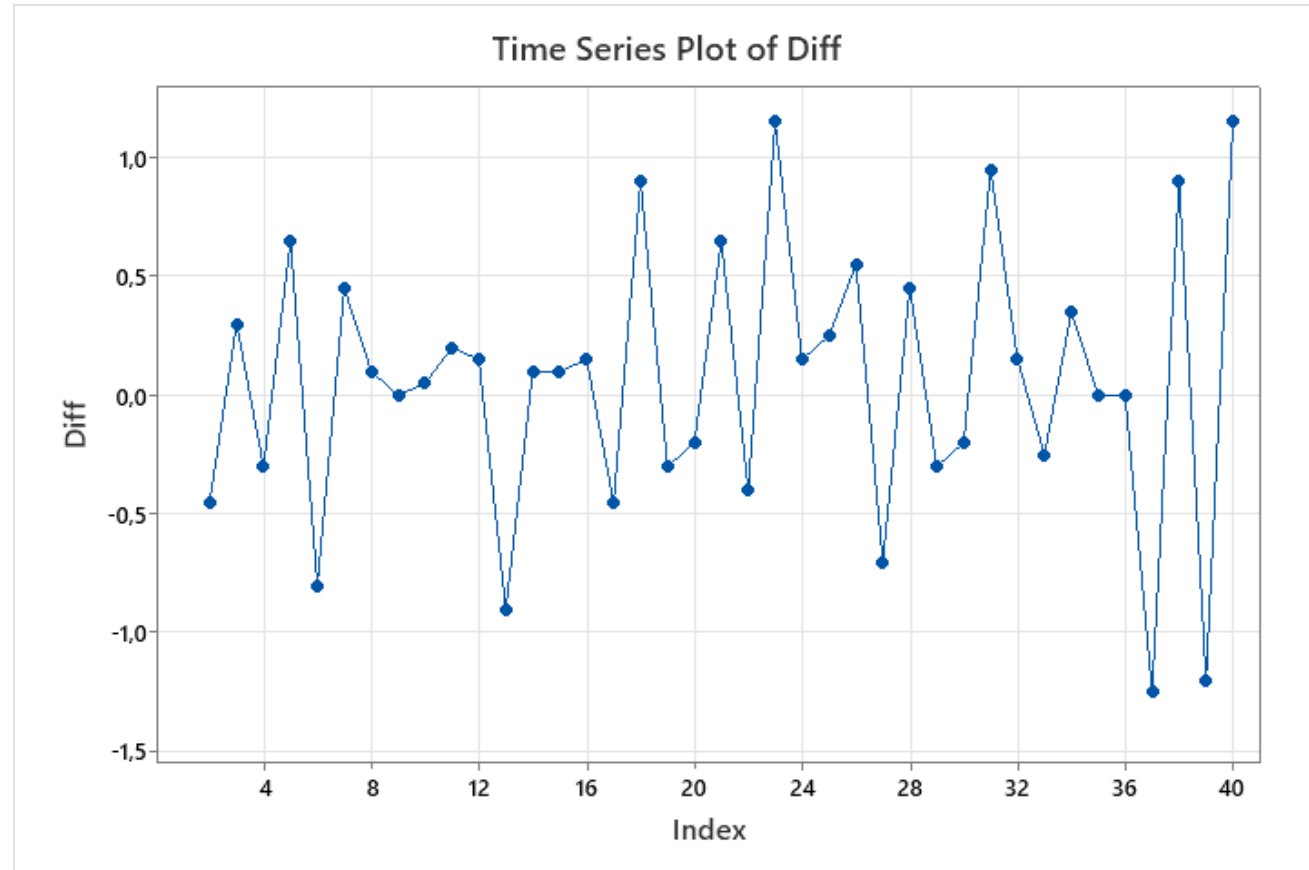
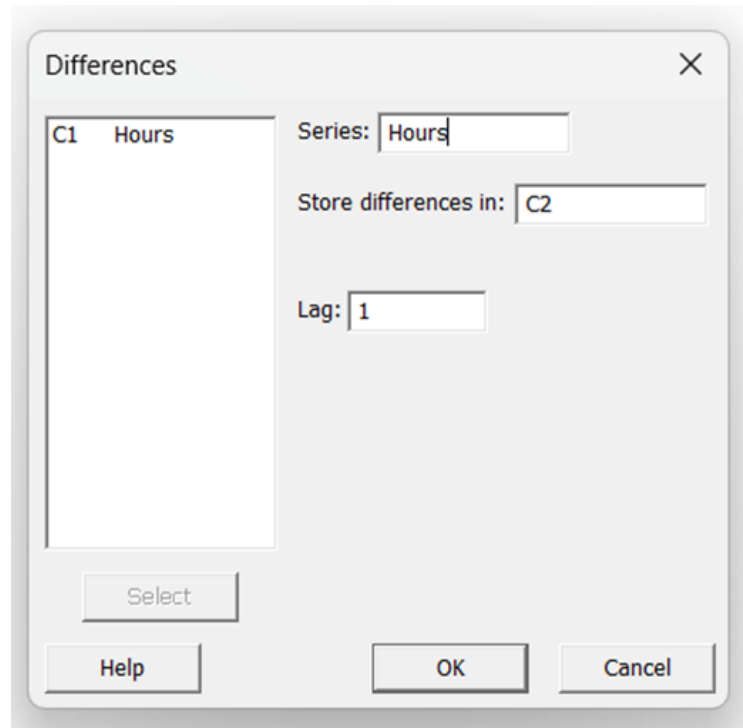
Number of Runs		
Observed	Expected	P-Value
2	20,80	0,000

The sample ACF follows a linear decay.
Non-stationarity needs to be addressed.

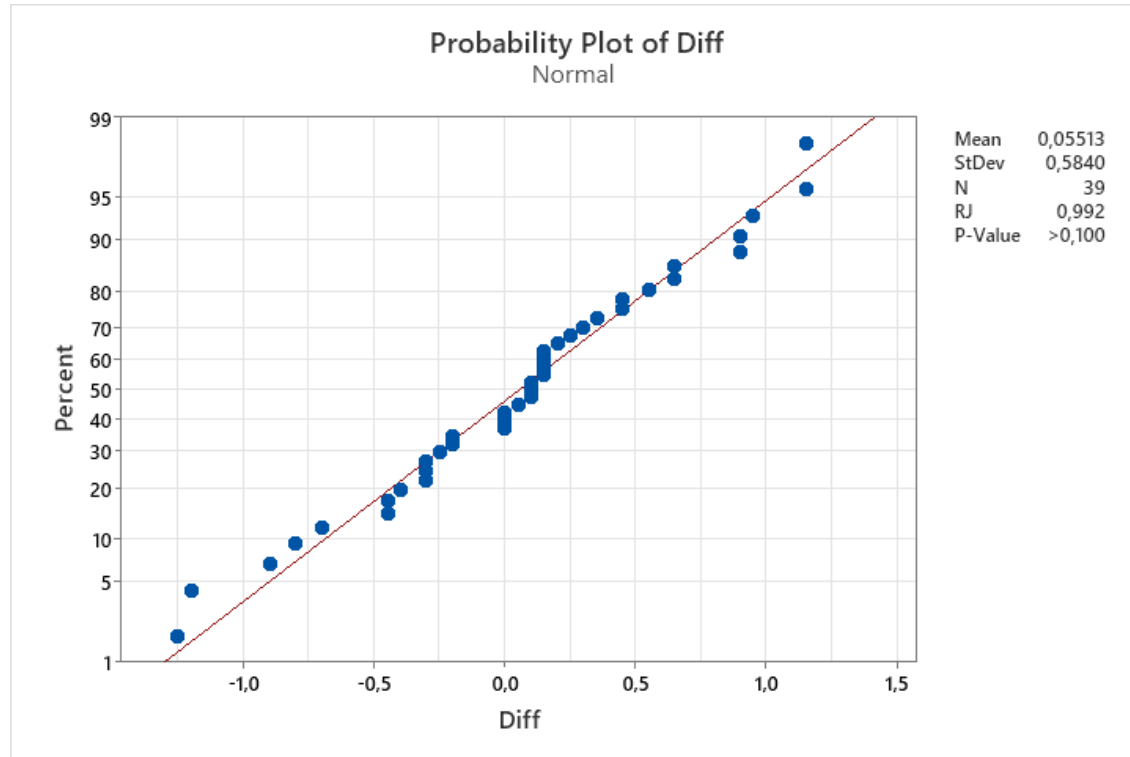
We **reject** the randomness

Point 2 Suggest a suitable model for the data.

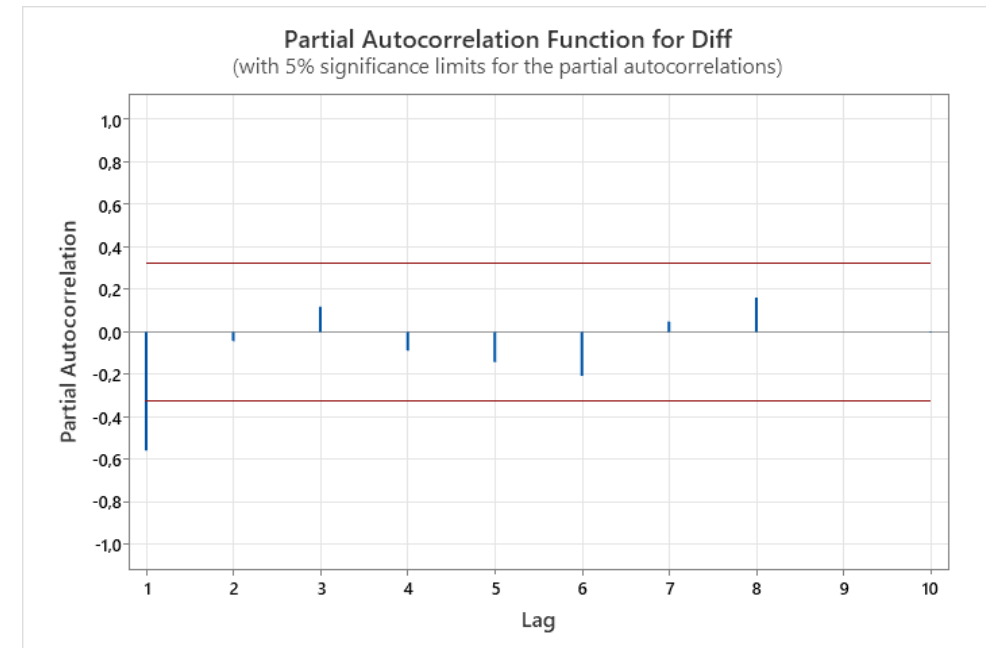
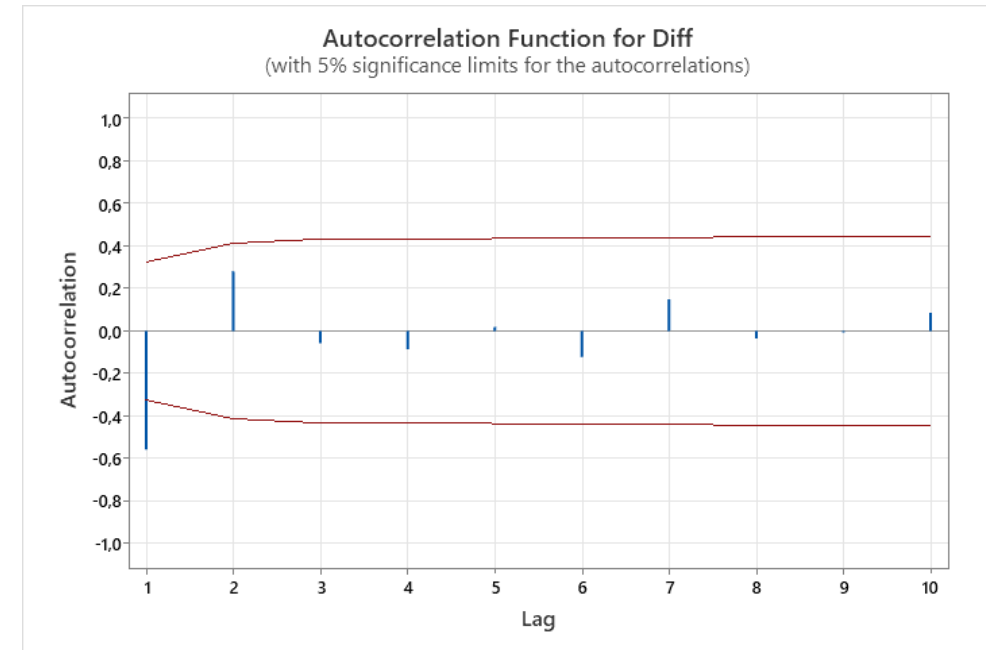
Let's try the **differencing operator**: *Stat → time series → differences*



Point 2 Suggest a suitable model for the data.



We can fit an **AR(1)** model to this new dataset (equal to ARIMA(1,1,0) on the original dataset).



Point 2

Suggest a suitable model for the data.

AR(1) model on the lag

Stat → time series → ARIMA

ARIMA

C1 Hours
C2 Diff

Series: Diff
Fit seasonal model
Period: 12

Autoregressive:

Nonseasonal: 1
Seasonal: 0

Difference:

Nonseasonal: 0
Seasonal: 0

Moving average:

Nonseasonal: 0
Seasonal: 0

☒ Include constant term in model

☐ Starting values for coefficients:

Select

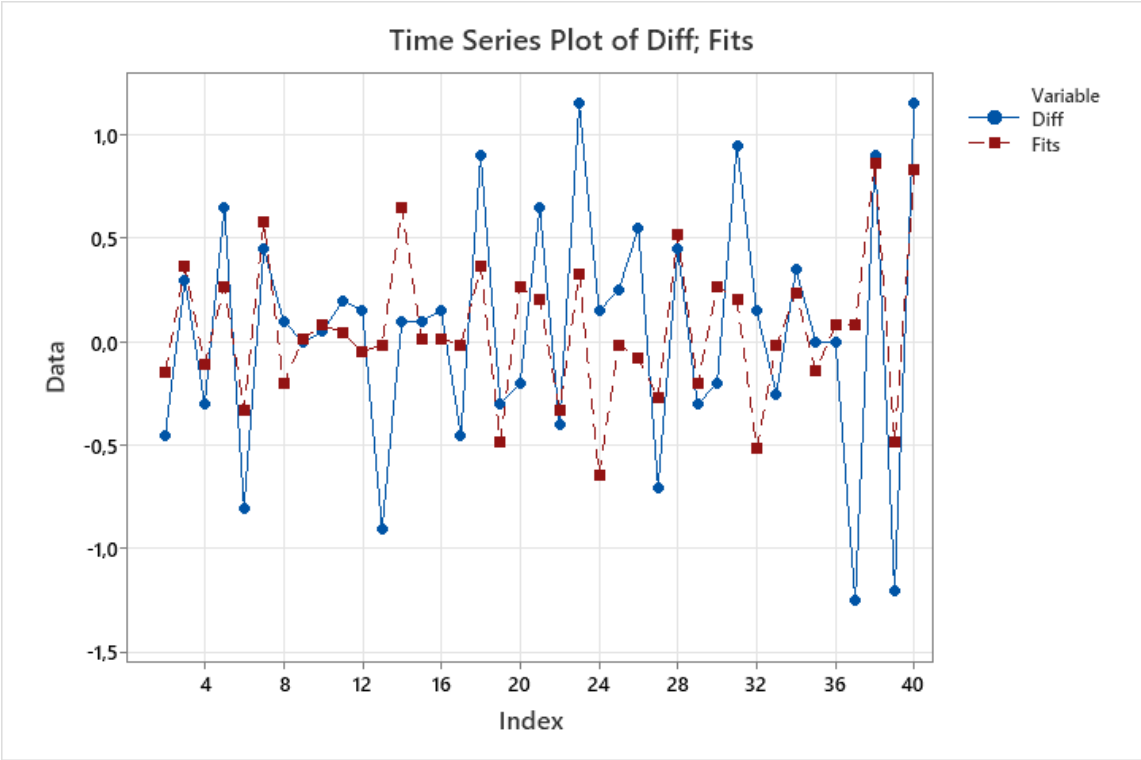
Graphs...
Results...
OK

Forecasts...
Storage...
Cancel

ARIMA: Storage

☒ Residuals
☒ Fits
☐ Coefficients

Help
OK
Cancel



Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	-0,628	0,138	-4,54	0,000
Constant	0,0800	0,0762	1,05	0,300
Mean	0,0492	0,0468		

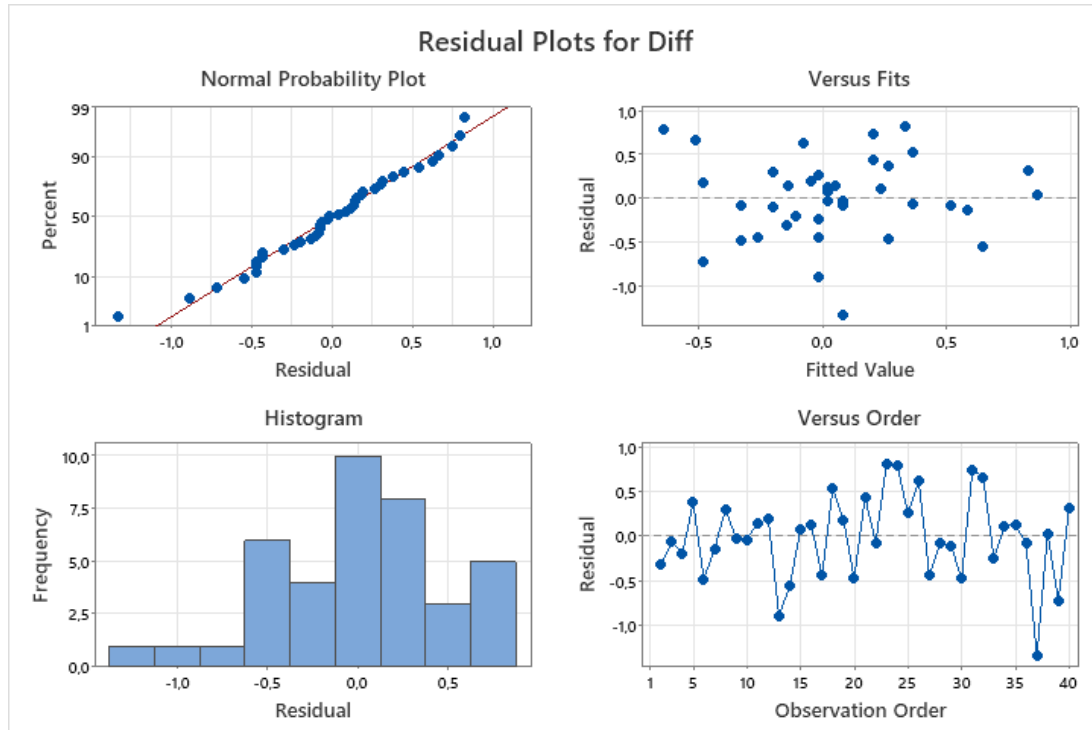
Regression equation:
$$\text{Diff} = + 0.0800 - 0.628 * \text{Diff_lag}$$

All the terms are significant.
Let's check if the **assumptions on the residuals** are met.

Point 2

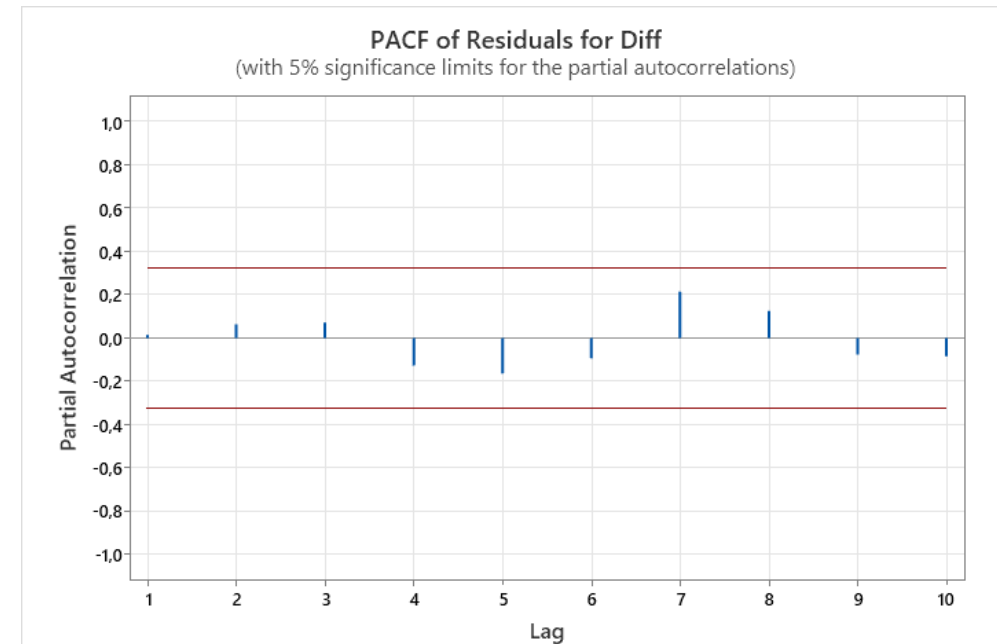
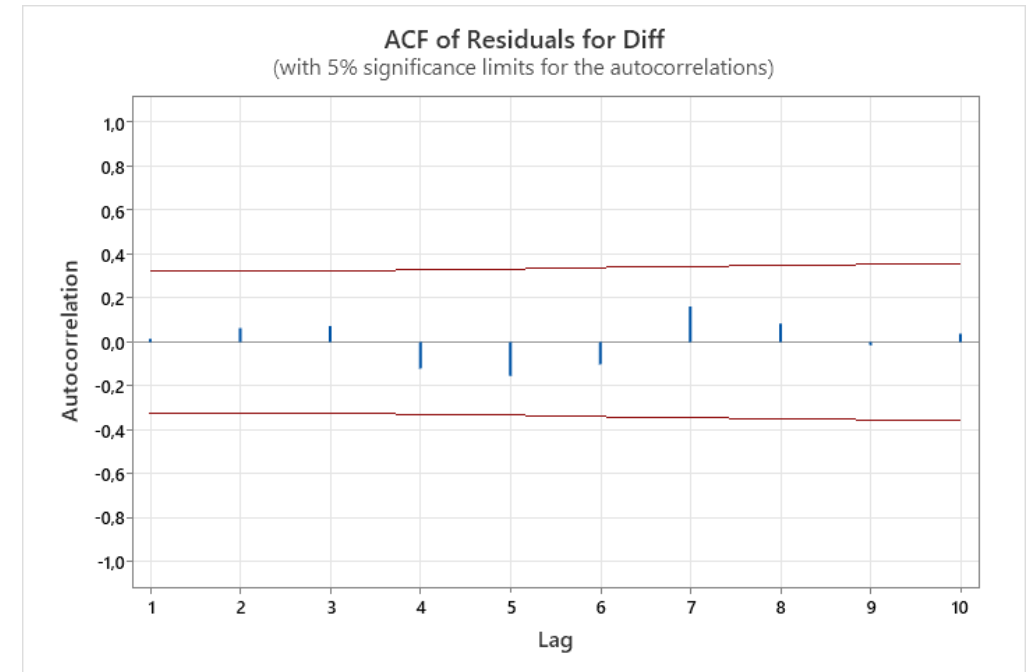
Suggest a suitable model for the data.

AR(1) model on the lag



- Runs test p-value on Residuals = 0,623
- Shapiro-Wilk p-value on Residuals > 0,1

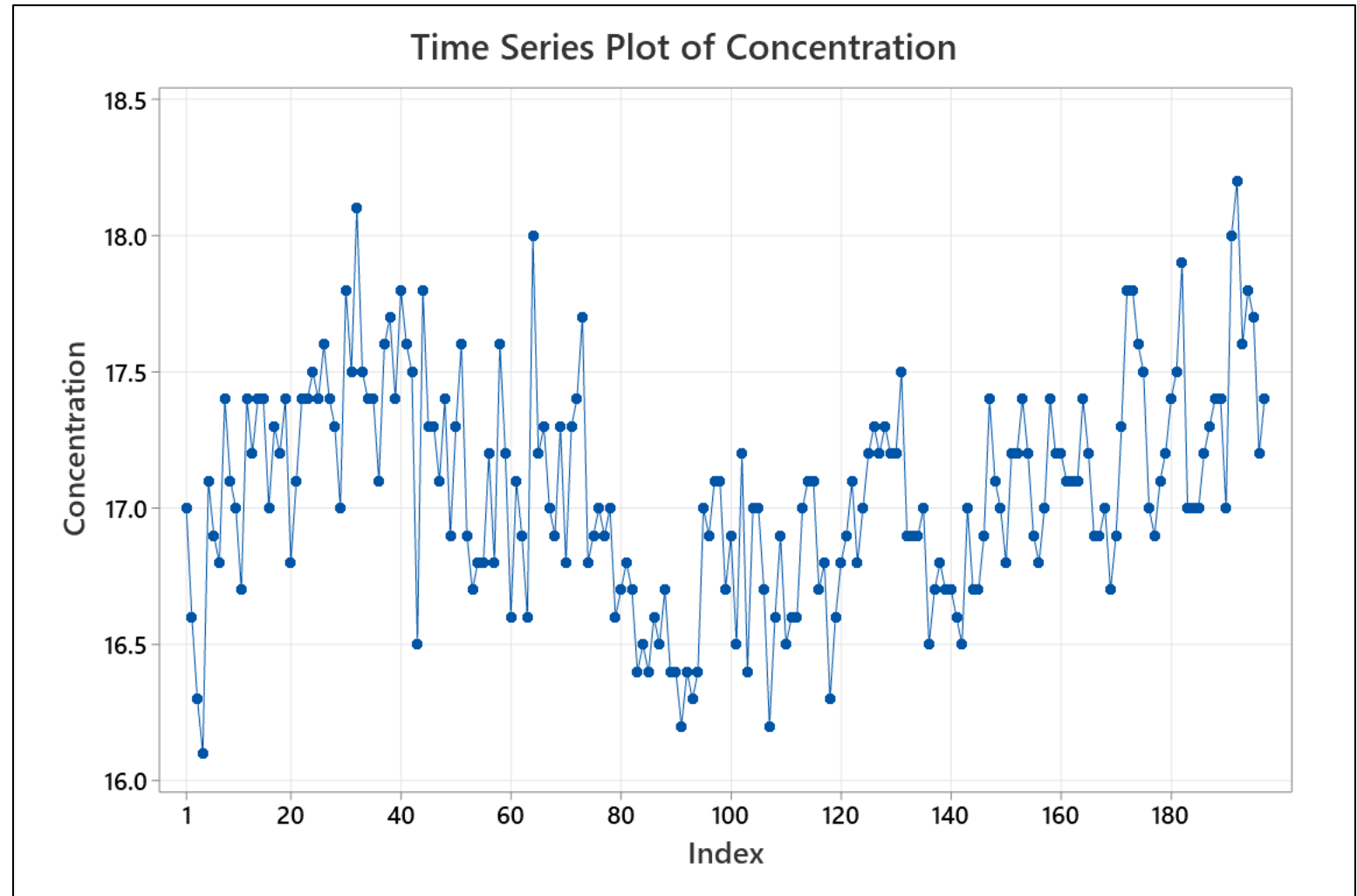
The model is adequate.



EXTRA EXERCISE 2

In a chemical process, data related to the concentration of a given component are measured every two hours. 197 consecutive observations are stored in `concentration.csv` (time series 'A' "Time Series Analysis – 3rd edition" Box Jenkins Reinsel – Prentice Hall)

1. Identify and fit a model for the data.



Point 1

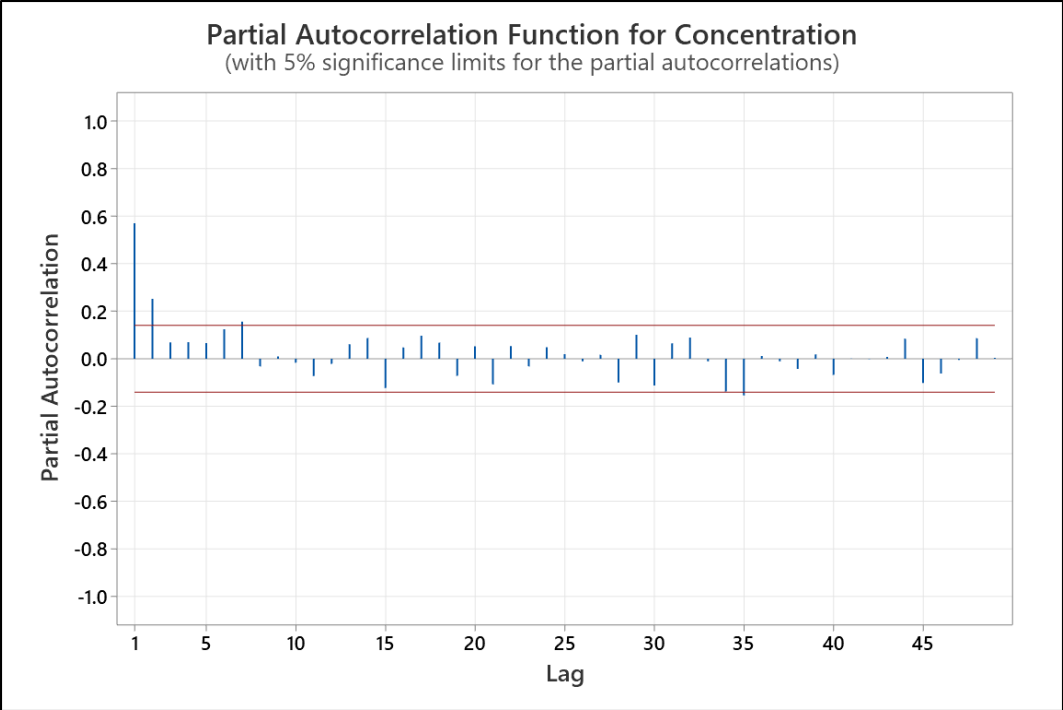
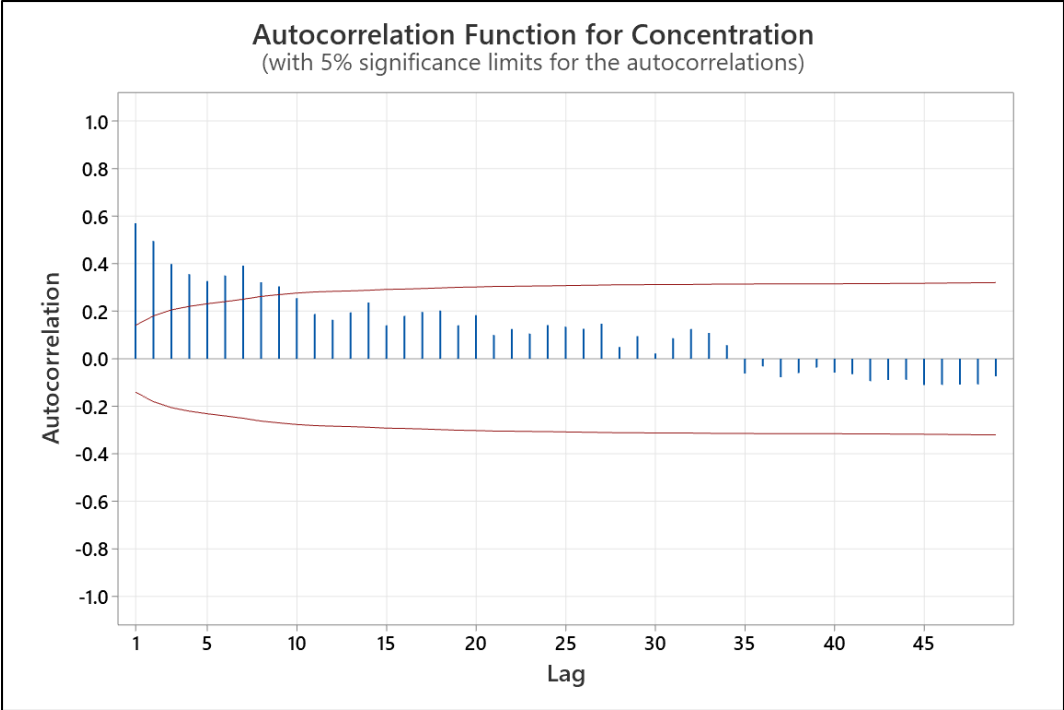
Identify and fit a model for the data.

Let's verify the independence assumption with runs test and ACF/PACF:

Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
52	99.48	0.000

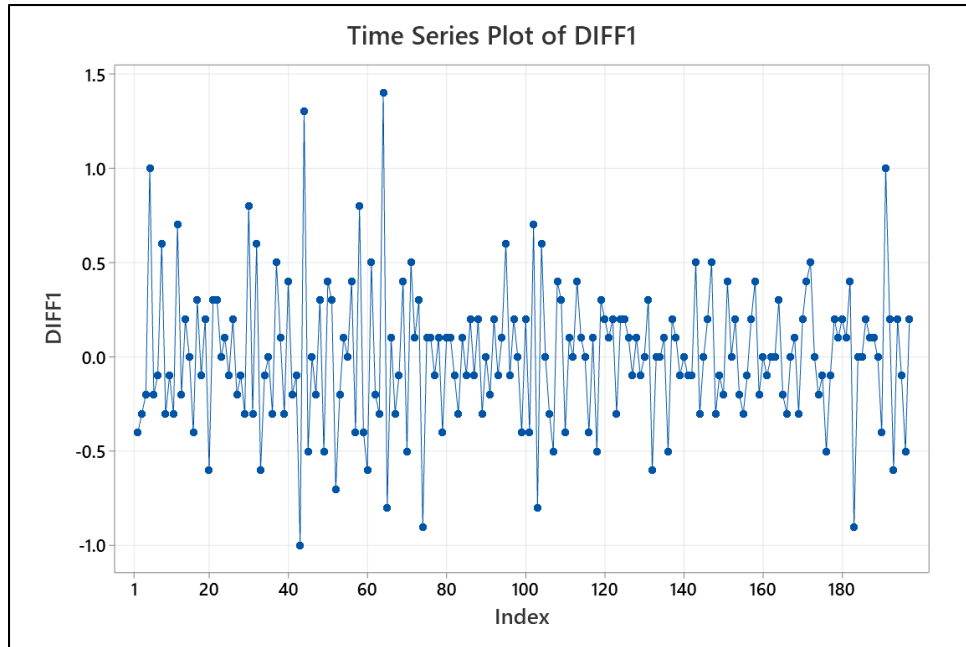


The process is **NON-STATIONARY**. Let's try to apply the difference operator.

Point 1 Identify and fit a model for the data.

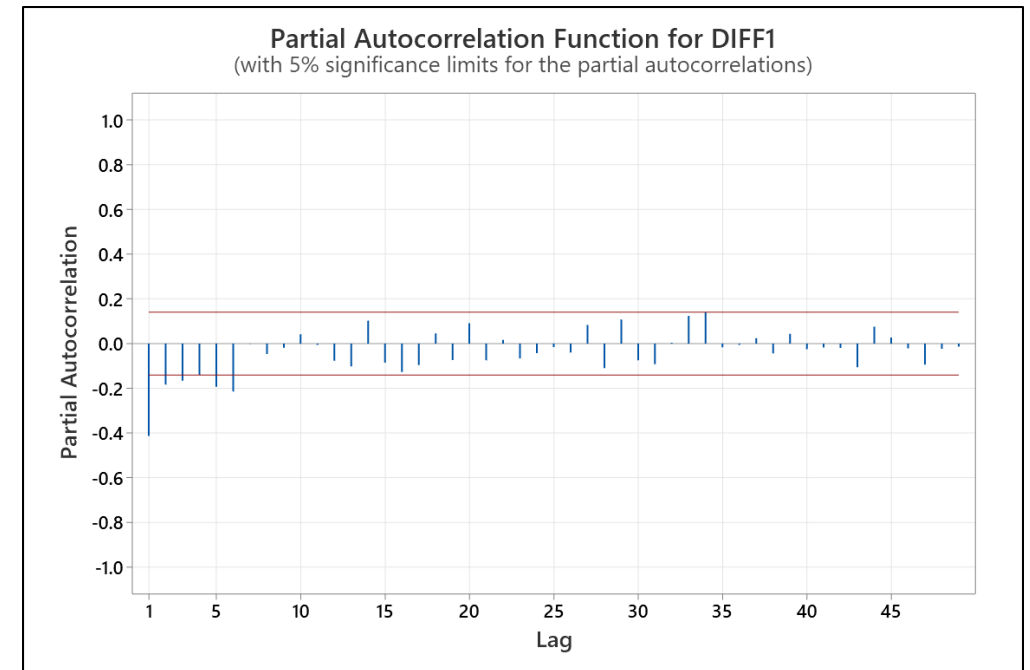
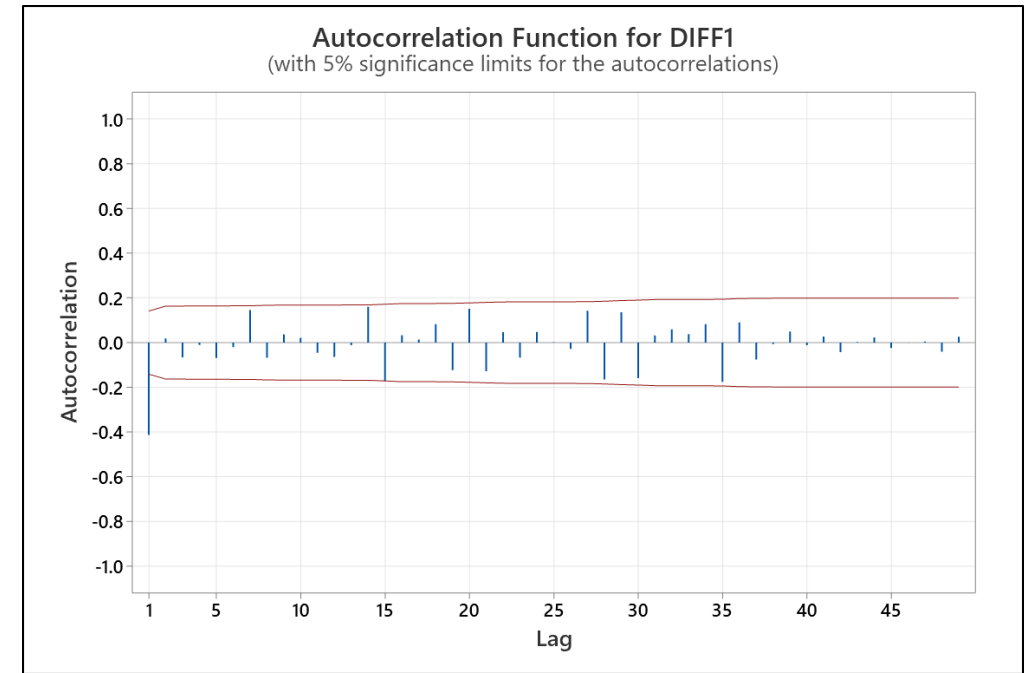
Let's try to apply the difference operator.

Stat → time series → differences



Runs test p-value on Diff = 0,00

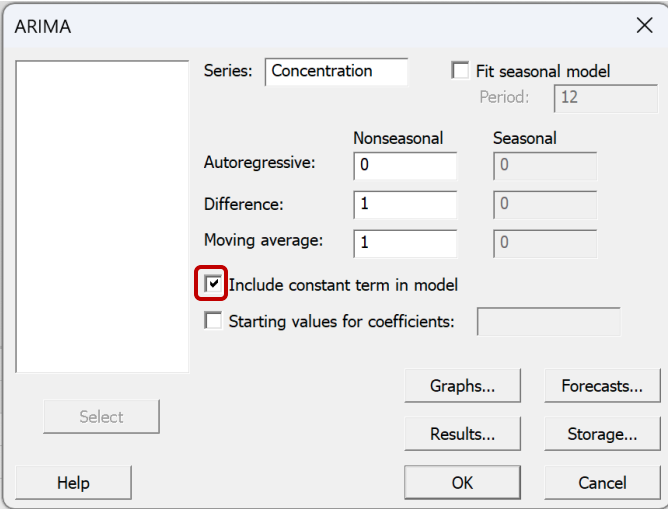
After the differencing operation, the most suitable model seems to be an MA(1). Thus, the investigated model is **ARIMA(0,1,1)**



Point 1

Identify and fit a model for the data.

Stat → time series → ARIMA



The ARIMA dialog box shows the following settings:

- Series: Concentration
- Fit seasonal model: ☐
- Period: 12
- Autoregressive: Nonseasonal 0, Seasonal 0
- Difference: Nonseasonal 1, Seasonal 0
- Moving average: Nonseasonal 1, Seasonal 0
- ☒ Include constant term in model
- ☐ Starting values for coefficients:

Buttons: Select, Graphs..., Forecasts..., Results..., Storage..., OK, Cancel, Help.

Final Estimates of Parameters

Type	Coef	SE	Coef T-Value	P-Value
MA 1	0.7100	0.0505	14.06	0.000
Constant	0.00405	0.00663	0.61	0.542

The computed ARIMA model is in the form:

$$Y_t - Y_{t-1} = \nabla Y_t = \mu - \theta_1 \epsilon_{t-1} + \epsilon_t$$

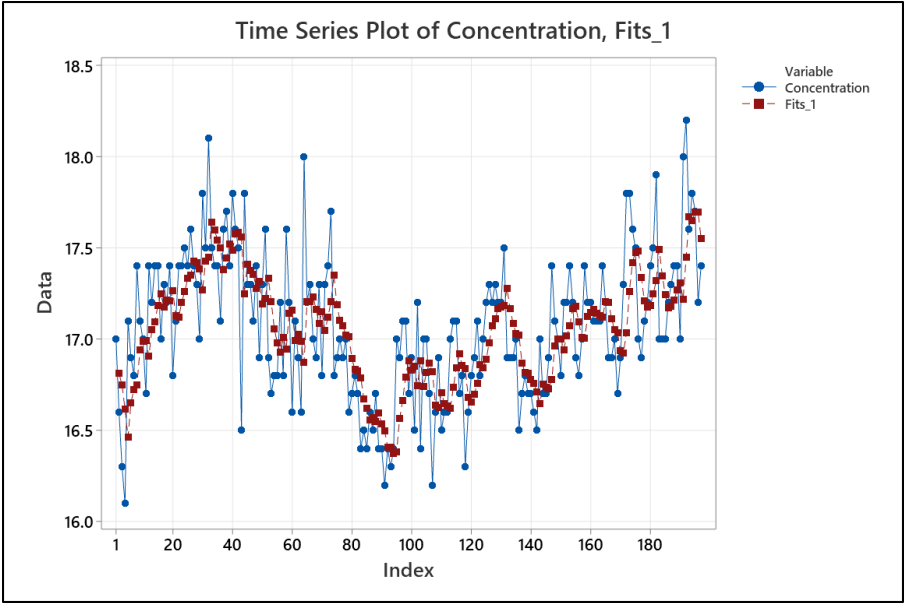
The constant term has a p-value of 0.542.
Let's remove the constant value by omitting the constant parameter.

Final Estimates of Parameters

Type	Coef	SE	Coef T-Value	P-Value
MA 1	0.7050	0.0507	13.90	0.000

The computed model is in now the form:

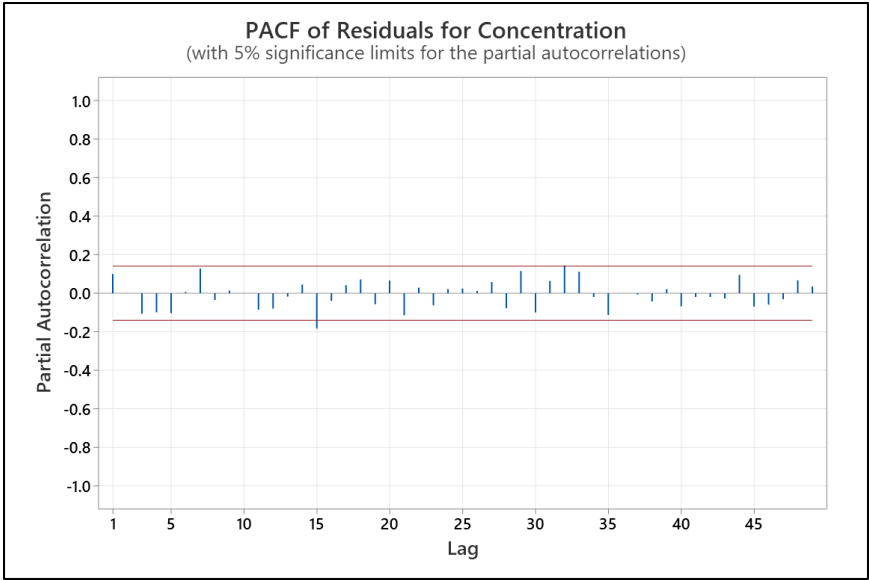
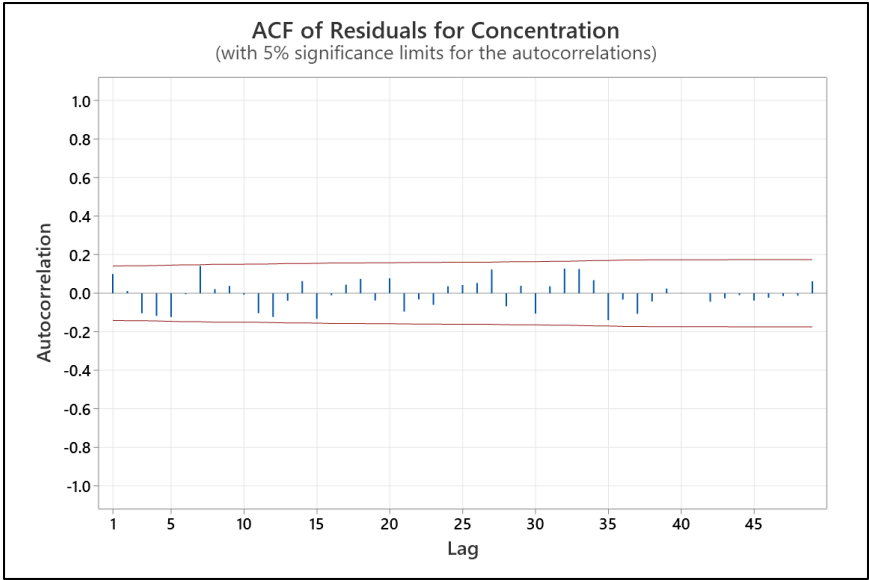
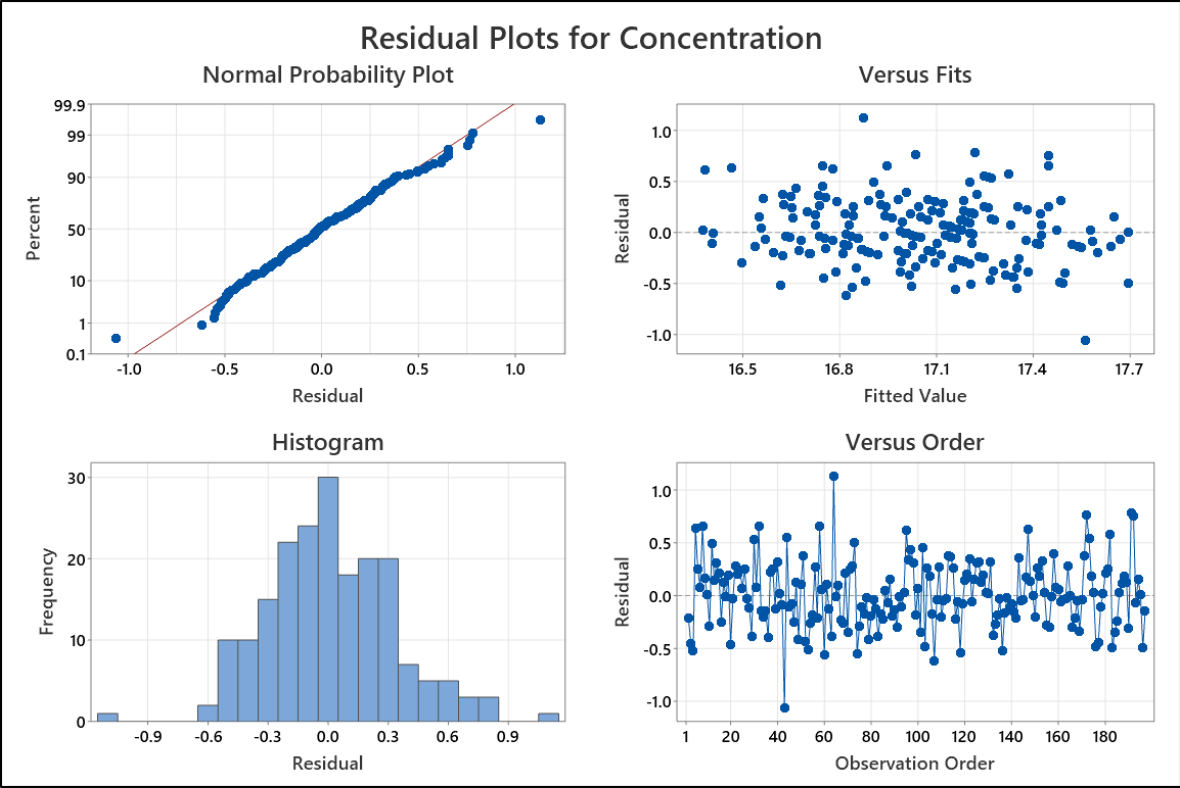
$$Y_t - Y_{t-1} = \nabla Y_t = \theta_1 \epsilon_{t-1} + \epsilon_t$$



Point 1

Identify and fit a model for the data.

Let's check the assumptions on the residuals:



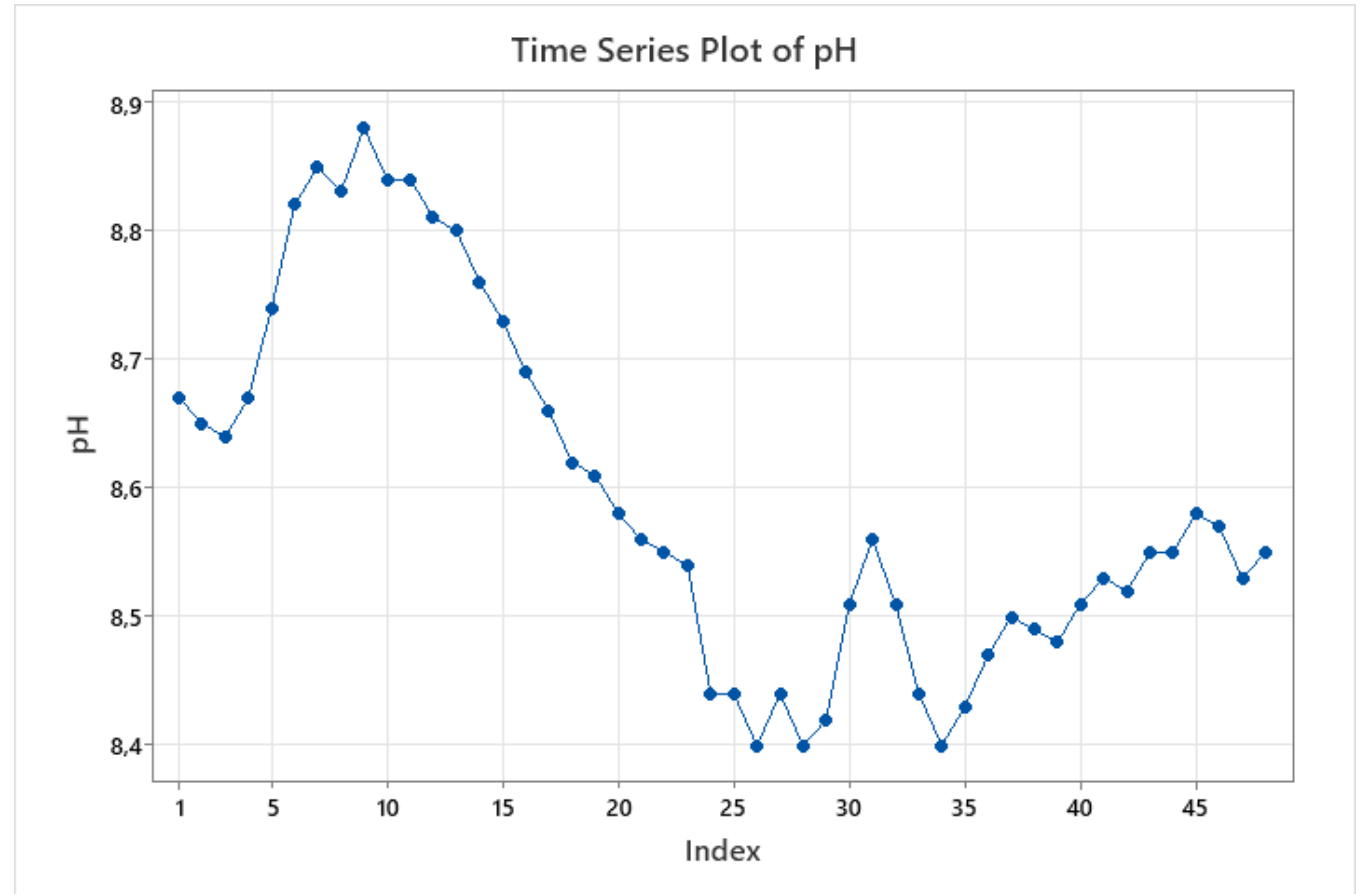
Shapiro-Wilk p-value on Residuals 0.072

The model is adequate.

EXTRA EXERCISE 3

In a chemical process it is necessary to keep constant the pH of a compound. Measurements are made every hour. Data acquired over the first 48 hours are reported in `pH.csv`.

1. Identify and fit a model for the data.



Point 1 Identify and fit a model for the data.

Let's verify the independence assumption with
Runs test and ACF/PACF:

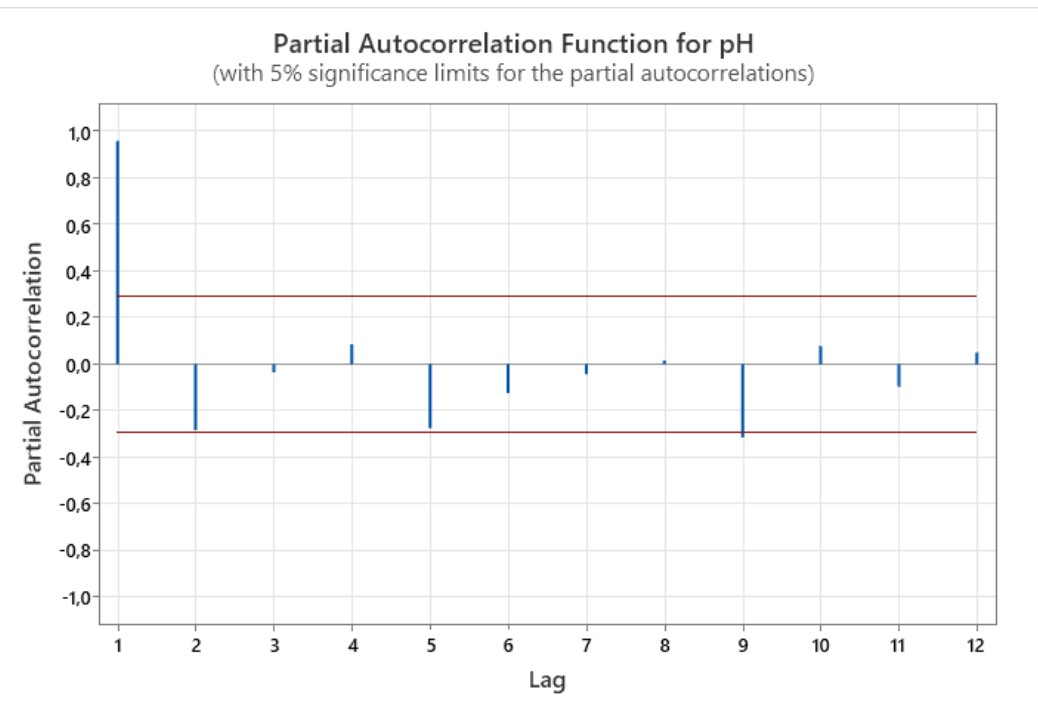
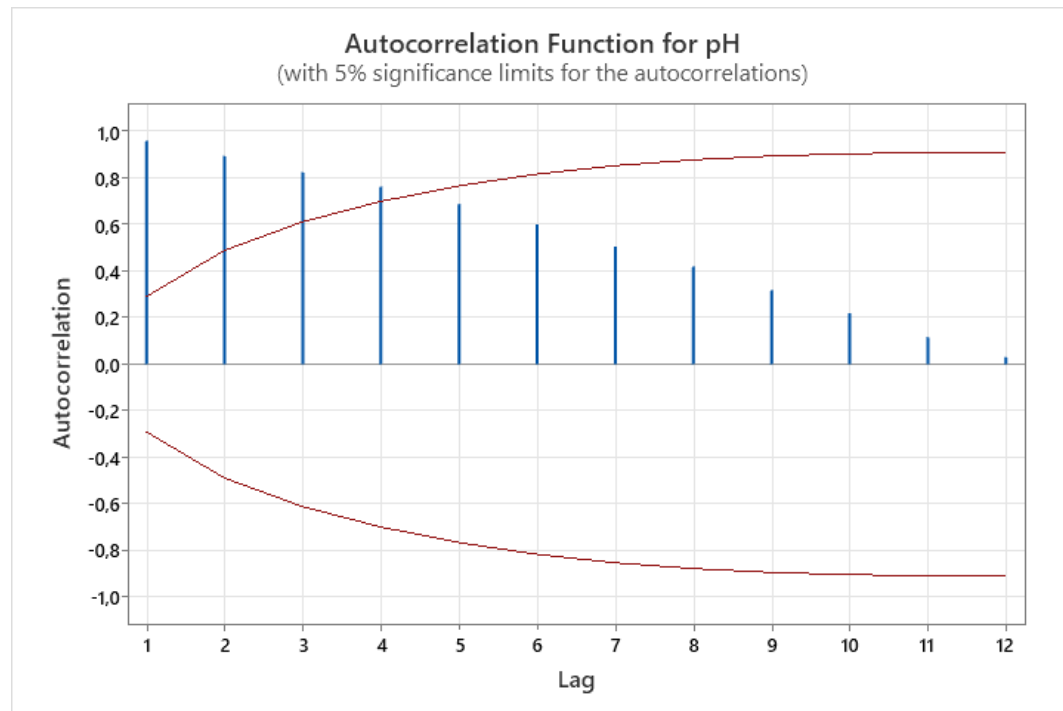
Test

Null hypothesis H_0 : The order of the data is random

Alternative hypothesis H_1 : The order of the data is not random

Number of Runs

Observed	Expected	P-Value
2	23,96	0,000



Data are not random. There is a strong positive correlation and decay of autocorrelation coefficients is not exponential. Based on ACF analysis, we can state that the process is **non-stationary**

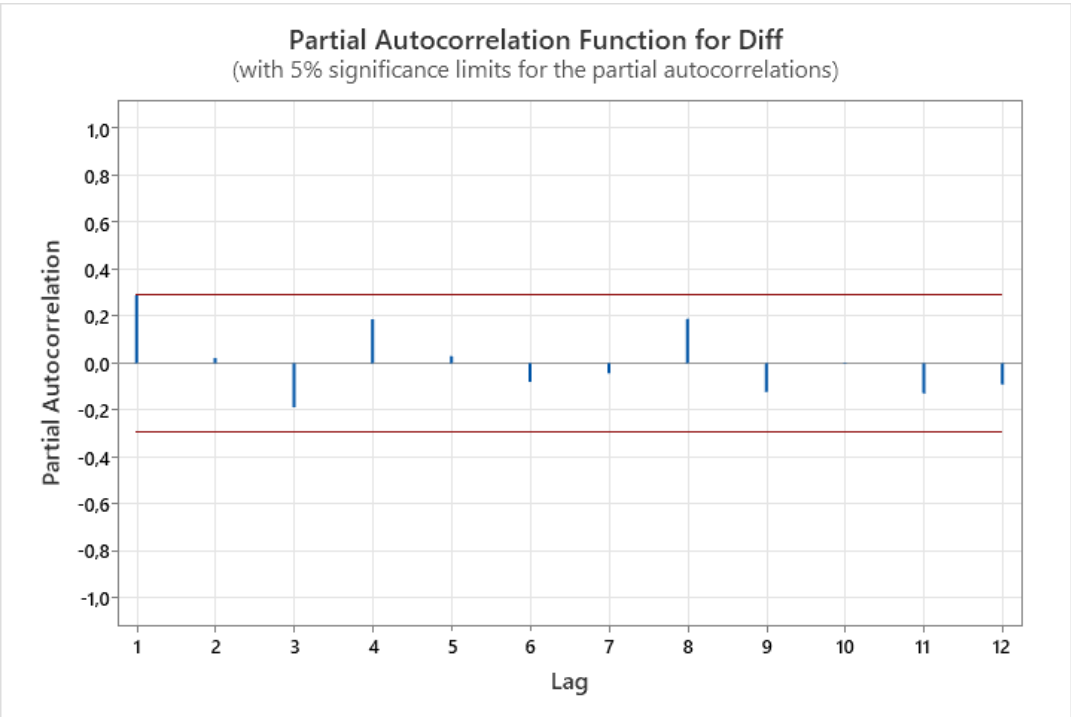
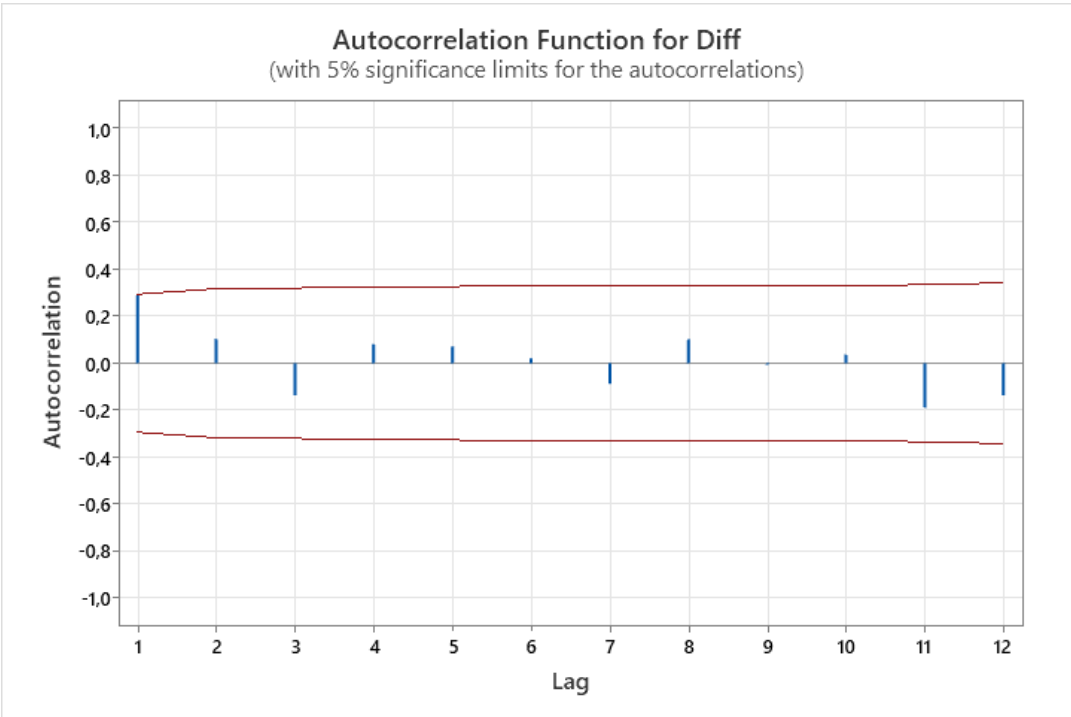
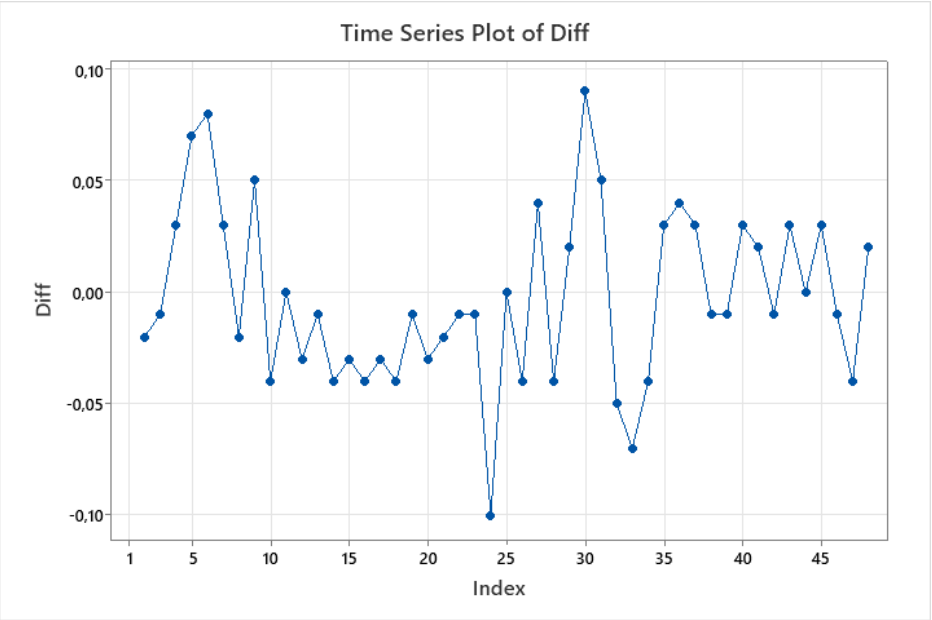
Point 1 Identify and fit a model for the data.

Let's apply the time difference operator to our time series data.

Runs test:

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
20	23,98	0,230

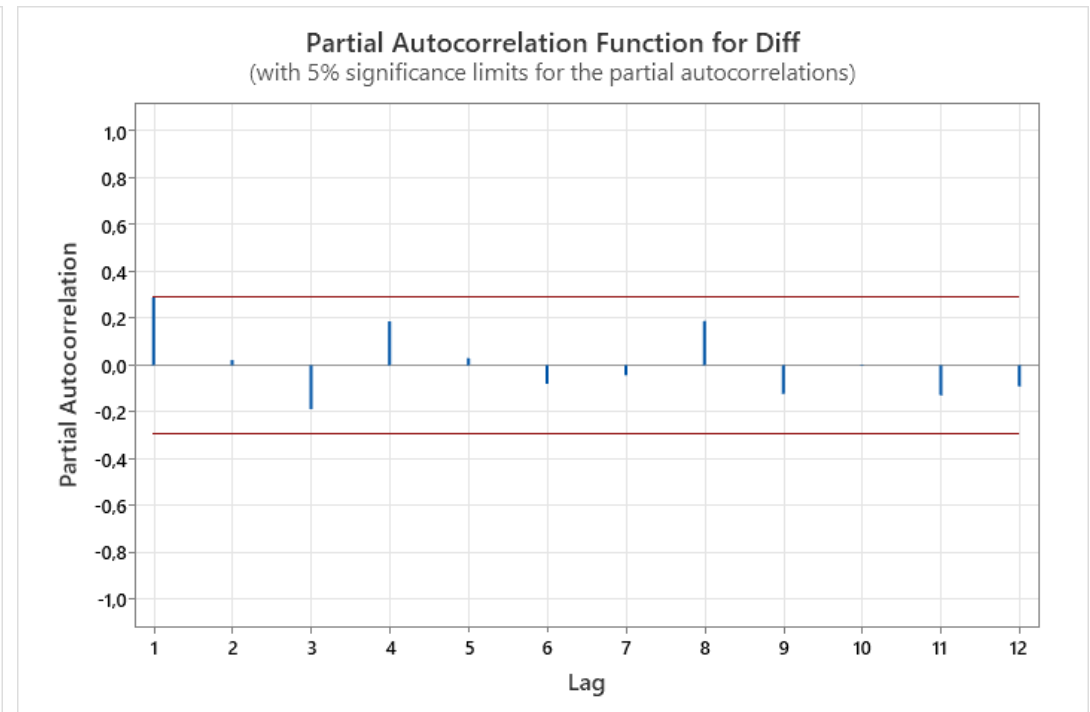
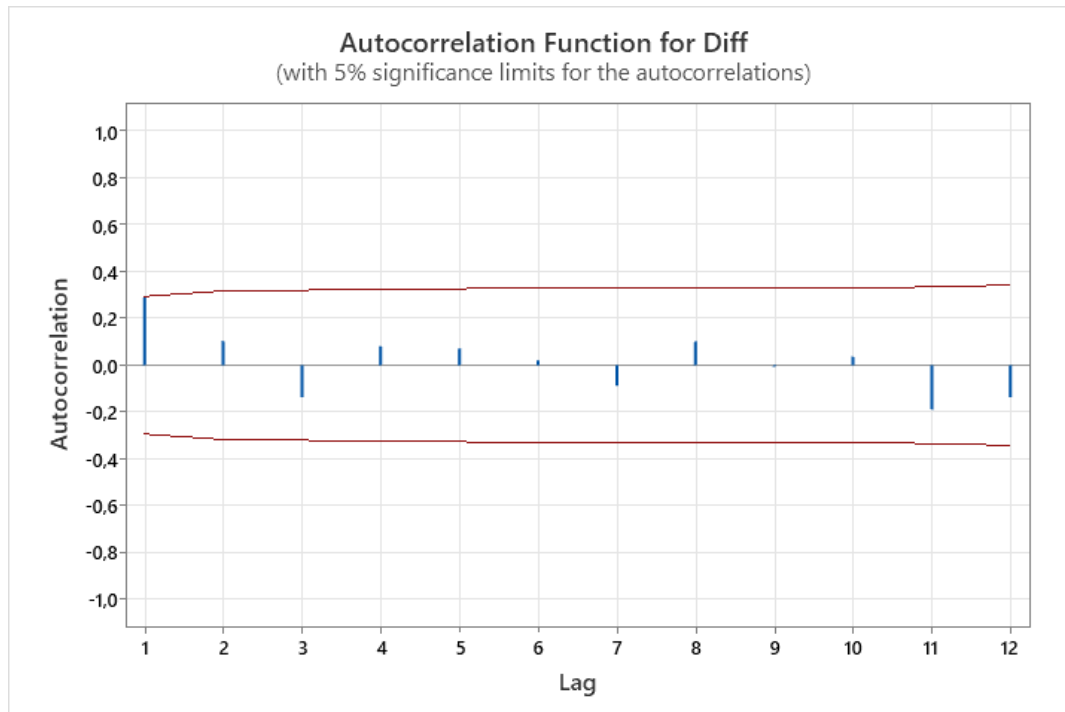


Point 1 Identify and fit a model for the data.

This is a typical borderline situation.

In this case, we may proceed by fitting an `AR(1)` model on the transformed data by applying the difference operator, and check if residuals are NID. If residuals are NID, the resulting model would be an **ARIMA(1,1,0)**.

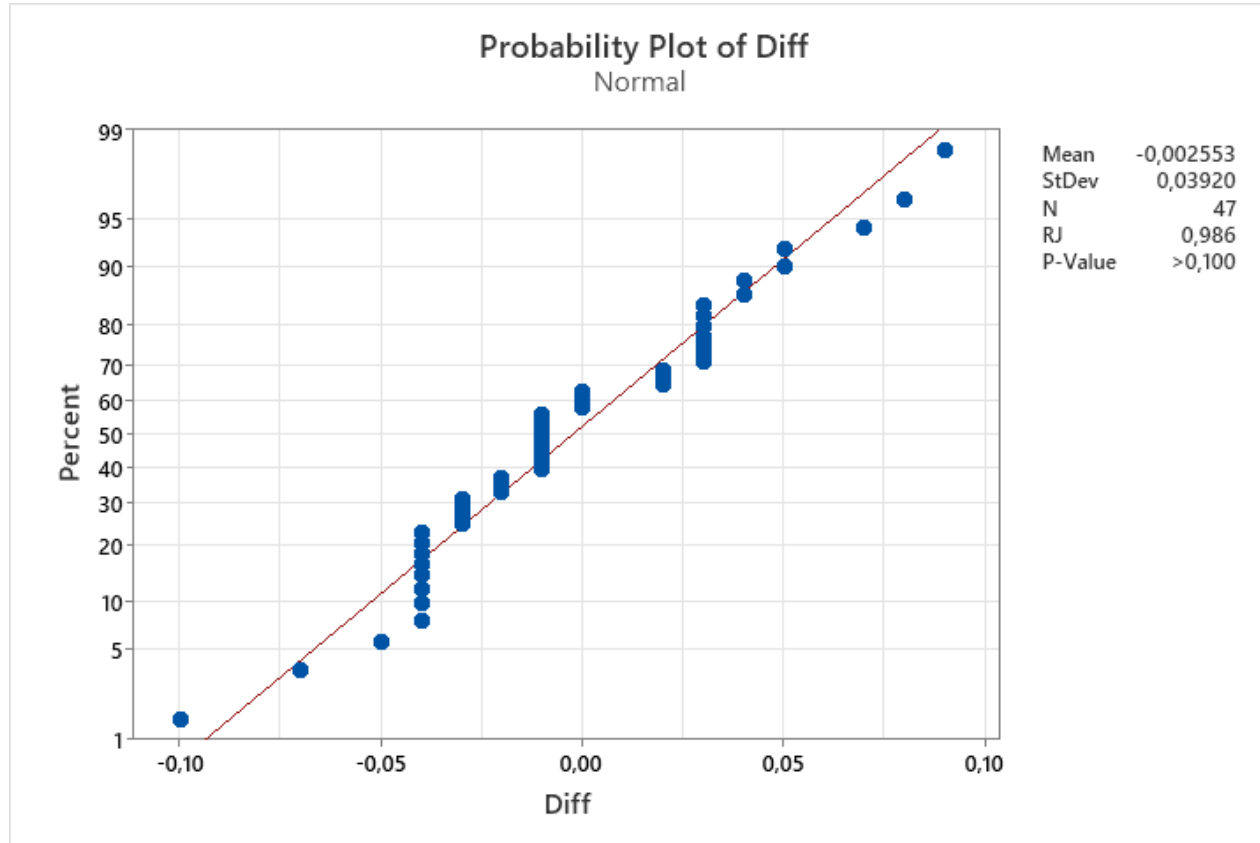
Another option is to **accept the transformed data are barely random**. If we follow this second path, we shall verify if they are also normal. Let's follow this second route.



Point 1

Identify and fit a model for the data.

Normality check:



The process is modeled as a **random walk**:

$$Y_t = Y_{t-1} + \epsilon_t$$

We can also try an **AR(1)** model directly on the original data (no application of the difference operator)

Point 1

Identify and fit a model for the data.

Stat → Timeseries → Lag

Specify series and lags:

Series	Lags
pH	1

Stat → Regresssion → Regression → Fit regression model

C1 pH

C2 Diff

C3 ACF1

C4 TSTA1

C5 LBQ1

C6 Residuals

C7 Fits

C8 pH_Lag1

Responses:

pH

Continuous predictors:

'pH_Lag1'

Categorical predictors:

Model...

Options...

Coding...

Stepwise...

Validation...

Graphs...

Results...

Storage...

Select

Help

OK

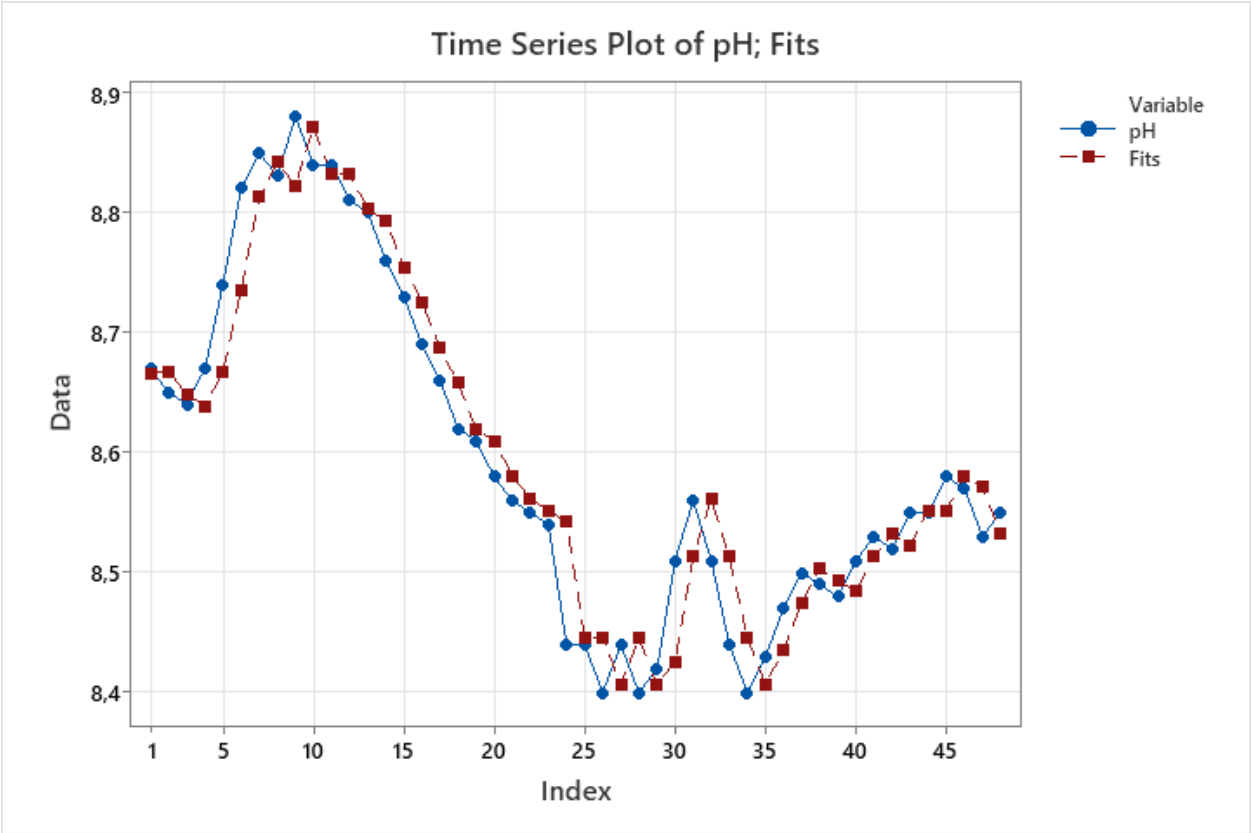
Cancel

Coefficients

Regression Equation

pH = 0,344 + 0,9597 pH_Lag1

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0,344	0,351	0,98	0,332	
pH_Lag1	0,9597	0,0408	23,51	0,000	1,00

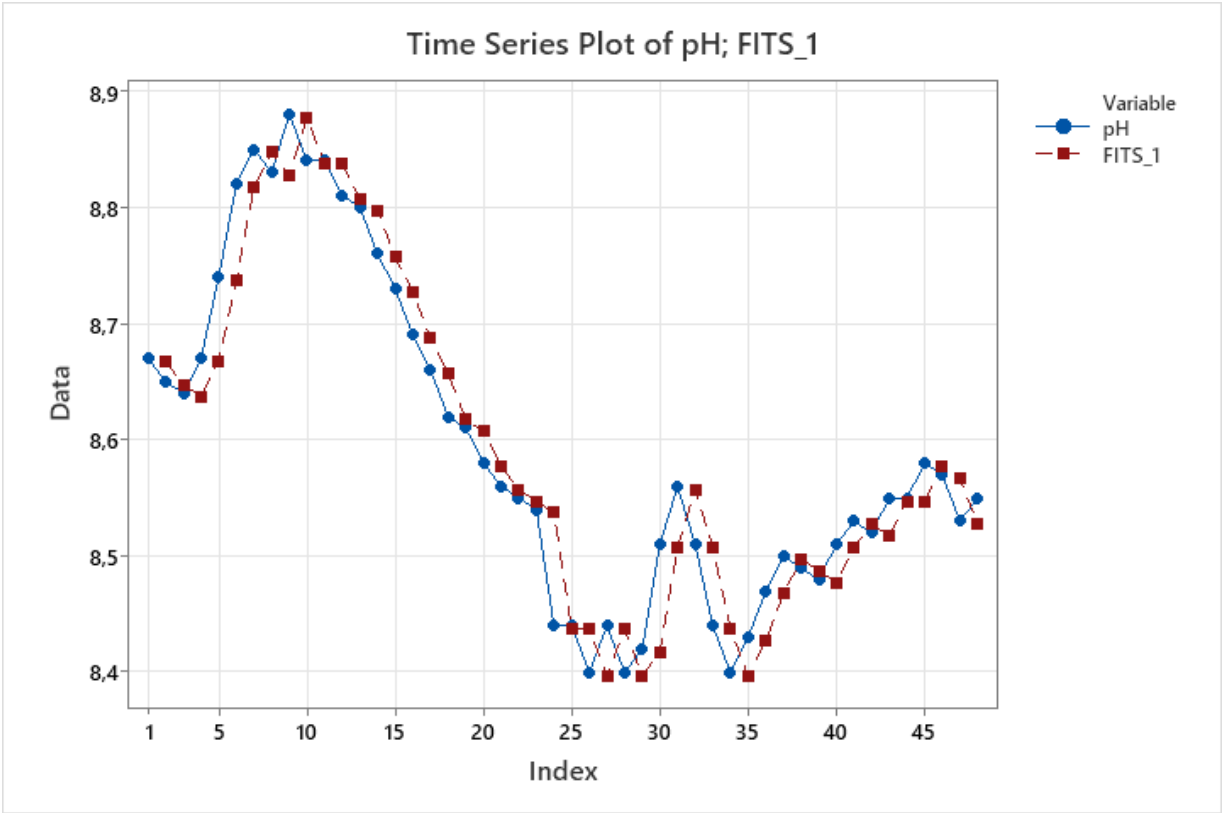
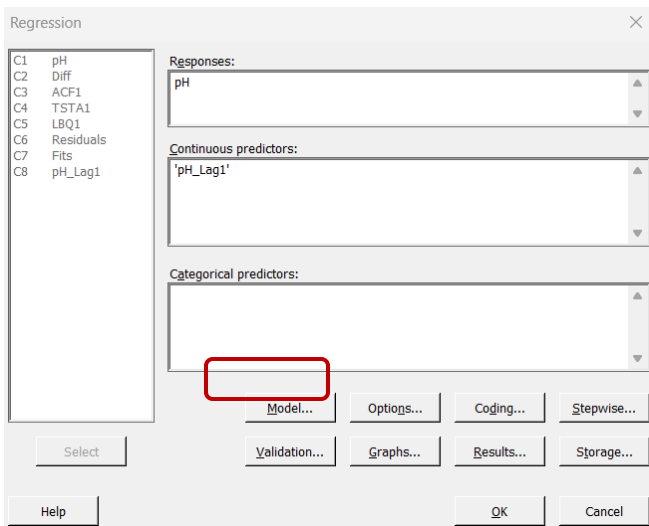


Let's re-calculate the model removing the constant term.

Point 1

Identify and fit a model for the data.

Let's re-calculate the model removing the constant term.



Regression Equation

$$\text{pH} = 0,999692 \text{ pH_Lag1}$$

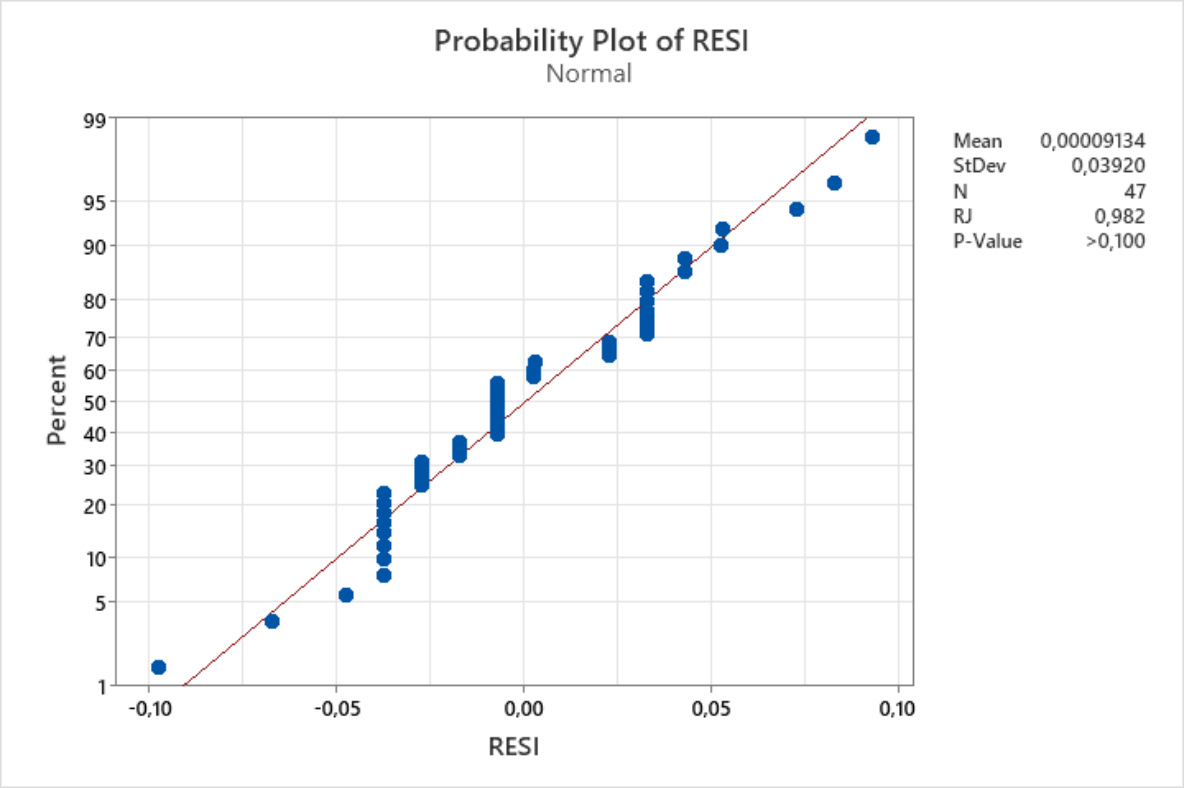
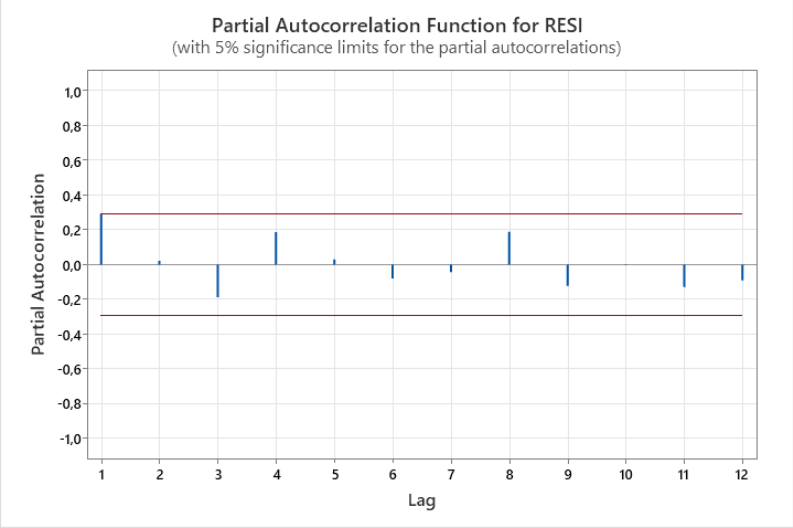
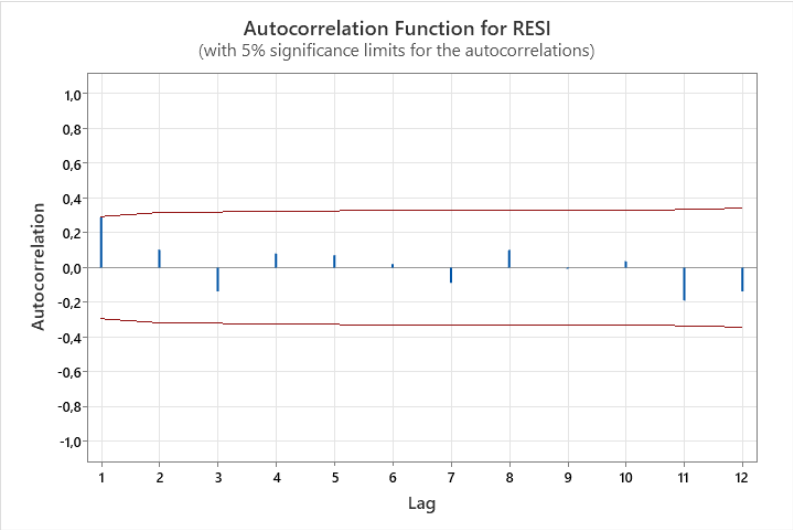
Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
pH_Lag1	0,999692	0,000665	1503,23	0,000	1,00

We have found again the random walk model!

Point 1 Identify and fit a model for the data.

Let's check assumptions on residuals



Test

Null hypothesis H_0 : The order of the data is random
Alternative hypothesis H_1 : The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
20	23,98	0,230

Same result as before
(indeed the model is the same,
i.e., random walk)